

GRNT APEX

Complete operating book

CPO sales deck, player→casino cascade, scenario math, Revolut partnership, Pay N Play rails, dual-mode wedge, binary CPMM, 500m lender, vAMM risk, and AMM vs exchange formulae. One file. Formula beats the deck.

Working name eGRIND is retired. Public product name is **GRNT APEX**. Figures are a model — not live P&L;, not a credit approval, not a quote. If a number in a slide conflicts with a formula here, the formula wins.

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1. Name, thesis, LTS

Candidate	Use	Verdict
GRNT APEX	Product + B2B widget	Primary.
GRNT TAPE	Screener / live feed	Module only.
GRNT RING	Consumer skin	Backup.
GRNT EDGE	Quant / EV engine	Engine, not the product.
eGRIND	Prior working title	Do not use externally.

Thesis. MMA fans will sit in a finance-grade terminal if v0.1 is paper money plus a route to a real book. Do not build an MGA book or an AMM until that conversion is measured.

Phase	Window	What ships	Money	Risk
1 Aggregator	Mo 1–3	Paper book, EV screener, copy feed, smart-route CPA. KPI: 500 users, >8 min, >15% CTR.	CPA 100–150 €	No licence, no float
2 B2B widget	Mo 4–12	White-label iFrame / SDK. Revolut Metal 50 € lock. Operator owns KYC.	SaaS from 6 k€/mo + 15% NGR	Operator licence
3 AMM Type 2	Mo 12–24	Own pool, liquidation. Mode A or B. Casino-wedge as production config.	Volume skim 70 bps	Licence + capital
4 Overlay	Y3+	Stream HUD, wallet ingress, multi-sport. DAZN stays here.	Rev-share / M&A;	Scale, not product

Casino wedge is sold from Phase 1 so the lawyer has an answer. It ships as production in Phase 3. Do not put DAZN, live AMM, or MGA Type 2 on the Phase 1 sprint.

Layer	Full vision	MVP 0.1
Cash	Fiat/crypto wallets	Virtual 10 000 €. Real stake via affiliate.
Risk	Live AMM + liquidation	Static margin formula on the client.
Odds	Monte Carlo HPC	Historical heuristic + The-Odds-API.
UX	In-book widget	Standalone PWA, dark terminal.
Social	Paid copy book	Public PnL board + 1-click copy of paper books.

2. CPO deck

APEX is the trading terminal the sportsbook is missing. Players already live in Robinhood and eToro. You still show them a 2010 coupon. APEX sits on your UI, turns a bet into a book, and routes the stake to you. Revolut Metal is the funded on-ramp. Plug-in widget — no backend rewrite. Phase 1 is paper + CPA.

Pain	Fact
Your product	Same Kambi/SBTech lines as every neighbour.
Your acquisition	Bonus war. CPA 150–250 €. Player leaves after the free bet.
Your tase	You are the counterparty on every slip. One-sided UFC card hurts.
Your session	~45 s. No book, no heatmap, no reason to stay.

The ask. Widget. Same 20x terminal. Dual-mode vault. GRNT senior revolver behind junior first-loss. Commercial: SaaS from €6 000/mo + 15% of APEX-routed NGR. Trustly/Zimpler/Brite stay on the cashier.

3. Player → casino feature map

Every player feature is only on the deck if it moves an operator KPI. Do not sell UX.

Player sees	Does	Casino gets	KPI
EV screener, true vs market, Kelly bar	Builds a book before staking. Stays on the card.	Session 45 s → 12–18 min	Session / touches
1x–20x isolated margin + liq band	Same deposit, more notional. Ruin capped to margin.	Volume velocity ~3.5x on unchanged deposits	Handle / GGR
Live risk band, open position not a coupon	Hedges and flash stakes during the fight.	More tx per event, less bounce after one slip	ARPU
1-click copy of verified paper books	Follows a quant instead of a bonus banner.	Organic invite loop. Lower CPA dependence	CAC / K-factor
Smart route to best combined price	One CTA after the book is built.	Inbound stake on the operator book	Conversion
Revolut Metal: 50 € locked margin	Opens a leveraged seat without a form.	Premium cohort, instant KYC, funded wallet	FTD / CAC
Same 20x UI in Mode A or B	No product change when licence rules bite.	Tase 0%. Host of the pool is a config flag	Variance / capital

4. Scenario math

Assumptions labelled. Not live evidence. Hold mix is an assumption — do not paste facility NPV into a first CPO mail. Legacy hold 4.5%. APEX hold 10 / 16.5 / 18% by scenario. NGR haircut 10%. GRNT take 15% of APEX NGR. SaaS €6 000/brand. CAC: legacy 180 €, APEX 95 €, Revolut Metal 55 €. Lock 50 €.

Input	Bear	Base	Bull
MAU	3 000	10 000	40 000
Deposit / MAU / mo	120 €	200 €	280 €
Leverage	2.0×	3.5×	5.0×
APEX hold	10%	16.5%	18%
Revolut share	8%	18%	28%
Brands	1	5	12

Output / month	Bear	Base	Bull
Deposits	360 000 €	2 000 000 €	11 200 000 €
APEX notional	720 000 €	7 000 000 €	56 000 000 €
Legacy GGR	16 200 €	90 000 €	504 000 €
APEX GGR	72 000 €	1 155 000 €	10 080 000 €
GGR multiple	4.4×	12.8×	20.0×
Casino keeps (after 15%)	55 080 €	883 575 €	7 711 200 €
GRNT NGR share	9 720 €	155 925 €	1 360 800 €
SaaS	6 000 €	30 000 €	72 000 €
GRNT month	15 720 €	185 925 €	1 432 800 €
GRNT ARR	188 640 €	2 231 100 €	17 193 600 €
Revolut users	240	1 800	11 200

Output / month	Bear	Base	Bull
Revolut float (50 €)	12 000 €	90 000 €	560 000 €
CAC saved vs legacy	264 600 €	922 000 €	3 848 000 €

Base case: 185 925 €/mo GRNT (2 231 100 € ARR), casino keeps 883 575 €/mo after the 15% share, GGR 12.8× legacy on the same deposits because leverage × hold. Hold mix is the swing factor — treat as a range, not a promise.

5. CPO matrix

Metric	Legacy	APEX	P&L;
UX	1X2 coupon	Leverage, heatmap, copy feed	Stops bonus-only acquisition
Hold	4–6% singles	15–25% packaged books	Only if parlays actually convert
Session	~45 s	12–18 min target	More touches per card
Tase	House vs player	AMM pool · Mode A or B	Wedge: host changes, curve does not
Integrate	Backend swap	iFrame / SDK	Days, not quarters
Cashier	Form + card	Existing PnP + Revolut Metal door	FTD without a form. Do not rip the rail
CAC	150–250 € bonus war	95 € APEX / 55 € Revolut Metal	Lower CPA, higher LTV (~3.8×)
Churn (mo)	~65%	~22% target	Session length is the leading indicator

6. Revolut Metal — partner, not a rail

Two different products. (1) Revolut as a bank inside Trustly/TrueLayer lists — already there, not the deal. (2) Metal hook for APEX: Open Banking-style seat, Metal KYC reused, **50 € locked as isolated margin**, instant on/off-ramp to the same balance. Needed for leveraged round-trips. Phase 2 on-ramp. Never sold as Pay N Play. Operator still owns the licence file.

Item	APEX treatment
What it is	Bank + wallet + 21–35 cohort. Partnership, not a cashier plugin.
KYC	Metal KYC reused. Not bank-login PnP.
Margin	50 € lock. Isolated. Ruin capped to the lock × lev.
Speed	Instant in and out to Revolut balance.
Fee	Interchange / FX. Not a 1% PSP take on GGR.
Phase	Phase 2. Phase 1 FTD rides the operator's existing Trustly.
Do not	Do not replace Trustly. Do not tell the CPO Revolut is Pay N Play.

7. Pay N Play rails

Figures as of 2026 public operator briefings — not quotes. APEX does not rip the cashier.

Rail	Fit	Markets / banks	Fee / contract	APEX
Trustly	Primary PnP	30+ EU. 6 300+ EU banks. Invented PnP 2016.	~0–1.2% in. High bar ~\$300k/mo, 12 mo.	Keep for FTD. Widget on top.
Zimpler GO	Alt PnP	SE, FI, DE, Baltics, BR Pix. TrueLayer closed Mar 2026.	~0.6–1.1%. Easier mid-market.	Second PnP if already on TrueLayer.
Brite Play	Alt PnP	~27 EU. 3 800+ banks. Fastest cash-out tests (~4 s).	~0.5–1.5% in / 0.5% out. ~\$200k/mo, 6 mo.	Pricing lever. Not first unless mid-market.
TrueLayer	Stack	UK + EU. Zimpler is the Nordic iGaming face.	Infra pricing.	If the UK book is already TrueLayer.
Nuvei	PSP	US+EU+LATAM. 720+ methods. Not PnP.	~1.5–3.5% + reserve.	Keep for cards. Not the FI identity rail.
Swish	SE only	Spelinspektionen sites.	Bank scheme.	Ignore unless SE-local.

Rail	Fit	Markets / banks	Fee / contract	APEX
Viljo / Trumo	Out	FI challenger, 5–10 casinos.	Unpublished.	Coverage risk. Do not build on it.
Revolut Metal	Partner	EEA+UK. Bank inside Trustly and a wallet.	Partnership.	Phase 2 on-ramp. Never a Trustly replacement.

If they have Trustly: keep it, put APEX on top, Revolut Metal as a second door for the leverage cohort. If new: Trustly for coverage, or Brite if the volume floor is the blocker. Never sell Revolut as Pay N Play. Never pull Nuvei as the Nordic identity rail.

8. Dual-mode wedge

The curve does not change. **Mode A:** GRNT hosts the vault (operator is a broker). **Mode B:** the same CPMM sits inside the licensed casino vault; the 500m facility is a hedge line, not a sweep of player funds. Sell Mode B to MGA/UKGC. Flip a flag. Tase 0% if the book is the pool and the house is not the taker.

Mode	Who holds cash	Who is counterparty	Sell to
A — GRNT hosted	GRNT vault	The pool. Operator is a broker.	Unlicensed / aggregator path
B — Casino vault	Licensed operator vault	The pool. Facility is a hedge line.	MGA / UKGC / local

9. AMM — five layers and the fill

L	Name	Job	Must not
L0	Senior facility	€500m revolving @ 6.5%, 7–30 day	Sit in one fight as k
L1	Junior first-loss	GRNT + operator. Halt lives here	Let senior pay winners
L2	Event line	Cap per fight / brand / group	Silo the same main 10 times
L3	CPMM book	$q_{yes} \cdot q_{no} = k$. Real cash layer	Set virtual L above the vault
L4	Isolated margin	Player posts m. Book sees m·lev	Cross-margin or portfolio net

Seed and buy YES (complete set)

$$q_{yes} = L \cdot (1 - p) \quad q_{no} = L \cdot p \quad k = q_{yes} \cdot q_{no}$$

$$\text{mid } p_{yes} = q_{no} / (q_{yes} + q_{no}) = p$$

$$q_{no}' = q_{no} + a \quad q_{yes}' = k / q_{no}'$$

$$\text{YES out} = a + (q_{yes} - q_{yes}') \quad p_{exec} = a / \text{YES out}$$

$$p_{quote} = \min(0.99, p_{exec} \cdot (1 + s)) \quad \text{odds} = 1 / p_{quote}$$

Worked fill — L = 10m, p = 0.62, a = 1.25m	Value
q_{yes} / q_{no} at seed	3 800 000 € / 6 200 000 €
k	23.560×10^{12}
YES out	1 887 584 € (a + swap 637 584 €)
p_{exec} (mid was 0.62)	0.6622 impact 6.8%
p_{quote} at 2% spread	0.6755 → odds 1.480
Cash to move mid +2%	166 428 €
Kyle $\lambda = 2(1-p)/L$	7.60e-08 per €
Same clip on OMX $\sqrt{Q}$ (Y=0.5, $\sigma=2\%$, ADV=80m)	0.12% AMM/equity $\approx 54\times$

€1.25m notional on a €10m line is a whale. Isolated 5x on €250k margin looks small to the player and large to the book. Cap notional against L, not against the deposit. λ overstates large clips (convexity) — use the exact CPMM for fills.

Inventory spread. Do not use 8h funding on a live fight.

$$s = s_0 + \kappa \cdot |\text{skew}| \quad \text{skew} = (q_{no} - q_{yes}) / (q_{yes} + q_{no})$$

$$\text{funding}_{8h} = f \cdot \text{skew} \quad (f = 8 \text{ bps at } |\text{skew}| = 1)$$

A fight is over in minutes. Funding is a skew tax between events, not a hedge during the round.

10. Facility, netting, isolated margin

$\text{monthly} = n \cdot H$ $\text{weekly} = \text{monthly} / (365/7/12)$
 $q_{\text{peak}} = \text{weekly} \cdot \text{peak_share}$ $R = q_{\text{peak}} \cdot (1 - s) / s$ $\text{need} = R / \text{target_util}$

Input / output	Base case
10 casinos × 40m handle / mo	400 000 000 € combined
Weekly / peak 8% one-sided	92 054 795 € / 7 364 384 €
R at 2% impact / need at 72% util	360 854 795 € / 501 187 215 €
Facility F / avg drawn 40%	500 000 000 € / 200 000 000 €
Interest 6.5% + commit 0.75%	15 250 000 € / year
LP take 0.70% of handle	33 600 000 € / year
Coverage LP / carry	2.20×
Event line 2% / group 4% / brand 15%	10 000 000 € / 20 000 000 € / 75 000 000 €
Siloed 10 books vs netting hub	100 000 000 € vs 20 000 000 € saved 80 000 000 €

€500m is the right senior size if $H \approx \text{€}40\text{m/month}$ per brand and peak is ~8% of weekly. If H is €80m+ or peak is 15% without a wider spread, add junior — do not enlarge the 6.5% loan. Tenor is 7–30 day revolving. Combat settles in hours. Not a five-year loan. Same UFC main on ten brands is one inventory. Siloed vAMMs multiply the one-sided hole.

Isolated margin: $m = N / \text{lev}$, $\text{lev} \in [1, 20]$. Book sees notional. Ruin capped to m . Liquidation unwinds into the CPMM; the house is not the taker. Queue >30s or gap >2%: reject new 10x+, ADL. Cross-margin forbidden. A parlay is a packaged book, not pooled margin. $\text{liq_odds} \approx o \cdot (1 - 1/\lambda)$ (haircut for spread/fees in production).

11. vAMM risk and halt tape

A pure vAMM sets k virtually and holds cash in a vault. Open interest can be all YES. Perpetual Protocol abandoned this and rebuilt v2 on real Uniswap v3 liquidity. A 15-minute fight is a worse vAMM than a BTC perp: funding cannot mean-revert before the bell, settlement is binary, recreational flow is structurally long the favourite.

Risk	House	Pure vAMM	Real CPMM	APEX
Counterparty	The house	Fund / protocol	Other token in k	Cash nets; $\text{lev} \leq \text{junior}$
Depth	Liability cap	Fake. $k > \text{vault}$	Slippage = cash	$L \leq 2\% F$ and $L \leq \text{vault}$
Funding	—	Hours–days	Not for solvency	Do not use on a fight
Settlement	House pays	Must return to pin	YES=1 from cash	Official result + ADL
One-sided	Tase spike	Core death	Price moves, backed	Spread, halt, group cap
Oracle	Own traders	Mark $\oplus$ oracle	Pool is the price	Halt if lag > 2s
Liquidation	Lose the stake	Cascade / bad debt	No lev in base	Isolated; 30s / 2% halt
Backstop	Equity	Print token	Complete-set	Junior. Senior freeze

Favourite-win hole (pure vAMM, all YES): $\text{margin} = N/\text{lev}$ $\text{payout} = N \cdot (1-p)$ $\text{uncovered} = \max(0, \text{payout} - (\text{vault} - \text{margin})) - \text{junior}$

Stressed example	Value
$N = 8\text{m}$ YES, $p = 0.62$, 5x, vault 3m, junior 1m	—
Margin posted	1 600 000 €
Winner payout $(1-p) \cdot N$	3 040 000 €
Uncovered	640 000 € HALT. Cut N , cut lev, or add first-loss. Do not widen k .

Halt tape

Code	Trigger	Action
SKEW	One side >70% of a market book	Widen spread, hike funding, invite other side
UTIL	Inventory >80% of the allocated line	Halt new leverage on that event
NAV	Pool NAV –5% day or –8% event	Kill switch. Junior absorbs. Senior frozen
LIQ	Liquidation queue >30s or gap >2%	Auto-deleverage, reject new 10x+
ORACLE	Odds lag >2s vs feed	Quote only, no fill
CORR	Same-direction flow across ≥ 4 brands	Group cap. Treat as one book
TENOR	Position open >24h	Force funding or close — facility is short
CAPACITY	Need > $1.05 \times F$	Widen s, cut lev, or add junior
CARRY	LP take / carry < $1.0 \times$	Raise skim or cut undrawn
VAMM	Uncovered hole > 0 on favourite win	Not a book. Reduce N or add first-loss

12. AMM vs exchange math

A CPMM is not a CLOB and not Betfair. Same €1.25m clip is a rounding error on OMX and a whale on a 10m fight line. Do not tell a CPO the fight book trades like a stock.

Desk calc	Exchange / CLOB / futures	APEX CPMM
Mid	(bid + ask) / 2, last, or index	$p = q_{no} / (q_{yes} + q_{no})$
Kyle λ	dP/dQ estimated from flow. Unstable	$\lambda = 2(1-p)/L$ at seed. Public
Impact	$I \approx Y \cdot \sigma \cdot \sqrt{(Q / ADV)}$ (Almgren)	$p_{exec} = a / YES_{out}$. Closed form
Depth	Shares inside 1 / 5 / 10 bps of mid	$a(i)$ that moves mid by i
VWAP	$\Sigma P_t Q_t / \Sigma Q_t$ vs the tape	The curve is the tape. No hidden size
Reservation	Avellaneda: $r = s - q \gamma \sigma^2 \tau$	$s = s0 + \kappa skew $. No time-to-close
Margin	SPAN / portfolio. Cross-net, VM daily	Isolated $m = N / lev$. No SPAN
Clearing	CCP novation, IM + VM	Complete set YES+NO = €1. Self-clears
Close	Auction, then T+1 / T+2	Binary 0/1 at the bell. No unwind to pin

	Equity CLOB	Betfair	Sportsbook	APEX
Who is the book	MM + the crowd	Back vs lay	The operator	The pool
A large clip	Walks the book	May rest unmatched	Limit or reject	Always fills. Price moves
Edge	Spread – adverse sel.	2–5% commission	Hold ~4.5–8%	Spread + IL + 70 bps skim
Leverage	Broker / futures	None in the match	Stake = max loss	Isolated 1–20x
Clock	Hours, then T+1	Until the event	Until the event	Minutes. Funding cannot save it

13. Operating rules (non-negotiable)

1. Never set virtual L above the vault on that line.
2. Never use funding as the solvency tool on a live fight.
3. Net the ten brands. Siloed vAMMs multiply the one-sided hole.
4. Cap OI so favourite-win payout $\leq$ junior. Senior does not pay winners.

5. Complete-set cash for unlevered flow. Leverage is bounded vAMM, not free k.
6. One event, one allocated line. Never one 500m Uniswap pool.
7. Mode A/B changes the legal host only. The curve is identical.
8. Phase 1 ships paper + CPA. Do not put DAZN, live AMM, or MGA Type 2 on the sprint.
9. Do not paste facility NPV into a first CPO mail. Hold mix is an assumption.
10. Do not replace Trustly. Do not sell Revolut as Pay N Play.
11. Do not tell a CPO the fight book trades like a stock.
12. If any halt code trips, stop new leverage. Do not wait through a main event.

Source of truth

Code: `src/lib/model.ts` (scenarios), `liquidity.ts` (CPMM), `lender.ts` (facility), `vamm-risk.ts` (hole), `pnp.ts` (rails), `exchange.ts` (λ and $\sqrt{Q}$). Interactive: APEX app — CPO deck, Lender pool, Pay N Play, Scenarios. Public read: github.com/isomarkkupentti/grnt-apex and github.com/isomarkkupentti/grnt-apex-playbook. Live snapshot: isomarkkupentti.github.io/grnt-apex. If the app and this PDF diverge, fix the PDF in the same commit as the code.

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